

PROJECT PORTFOLIO

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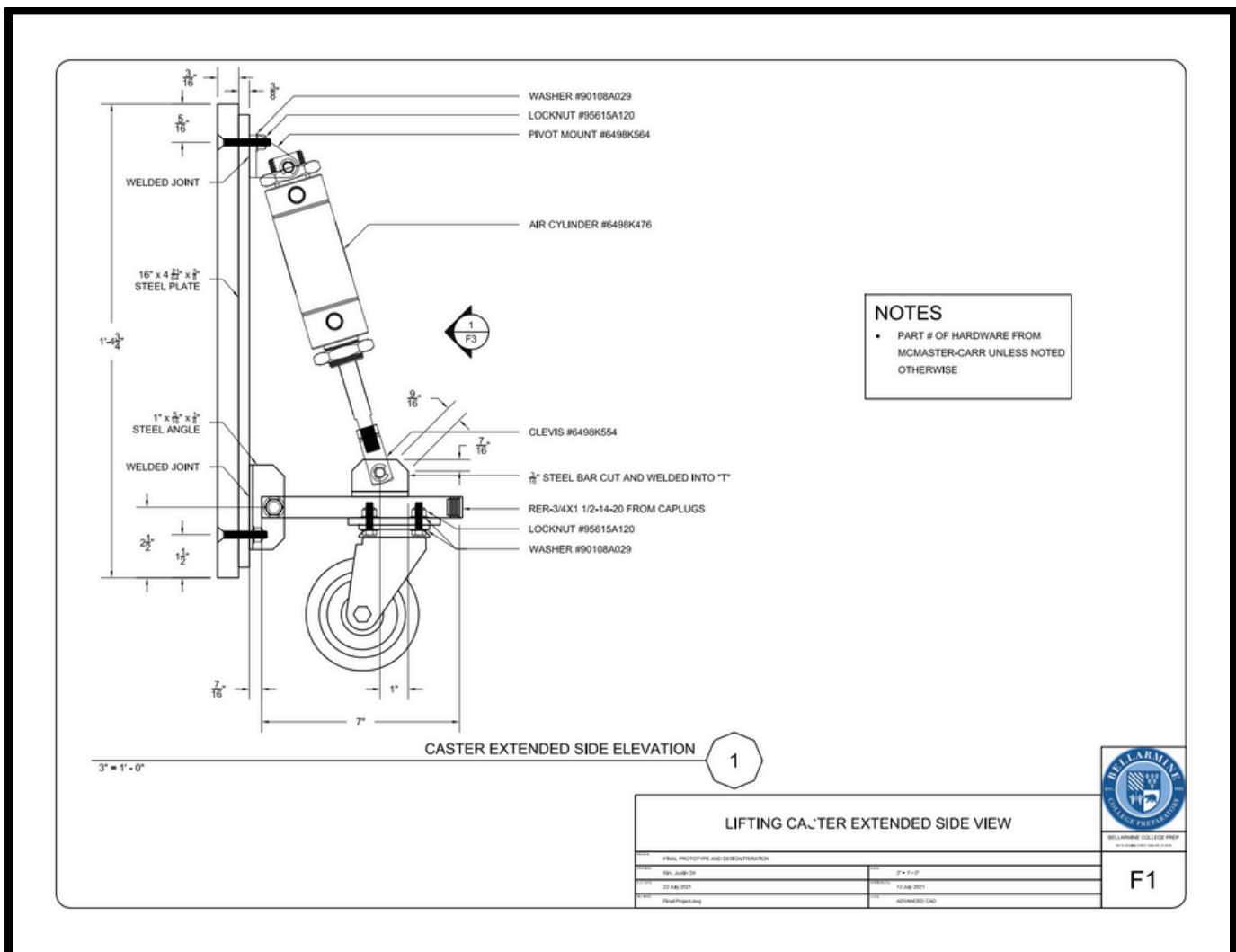
LIFTING CASTER (AutoCAD)

Summary

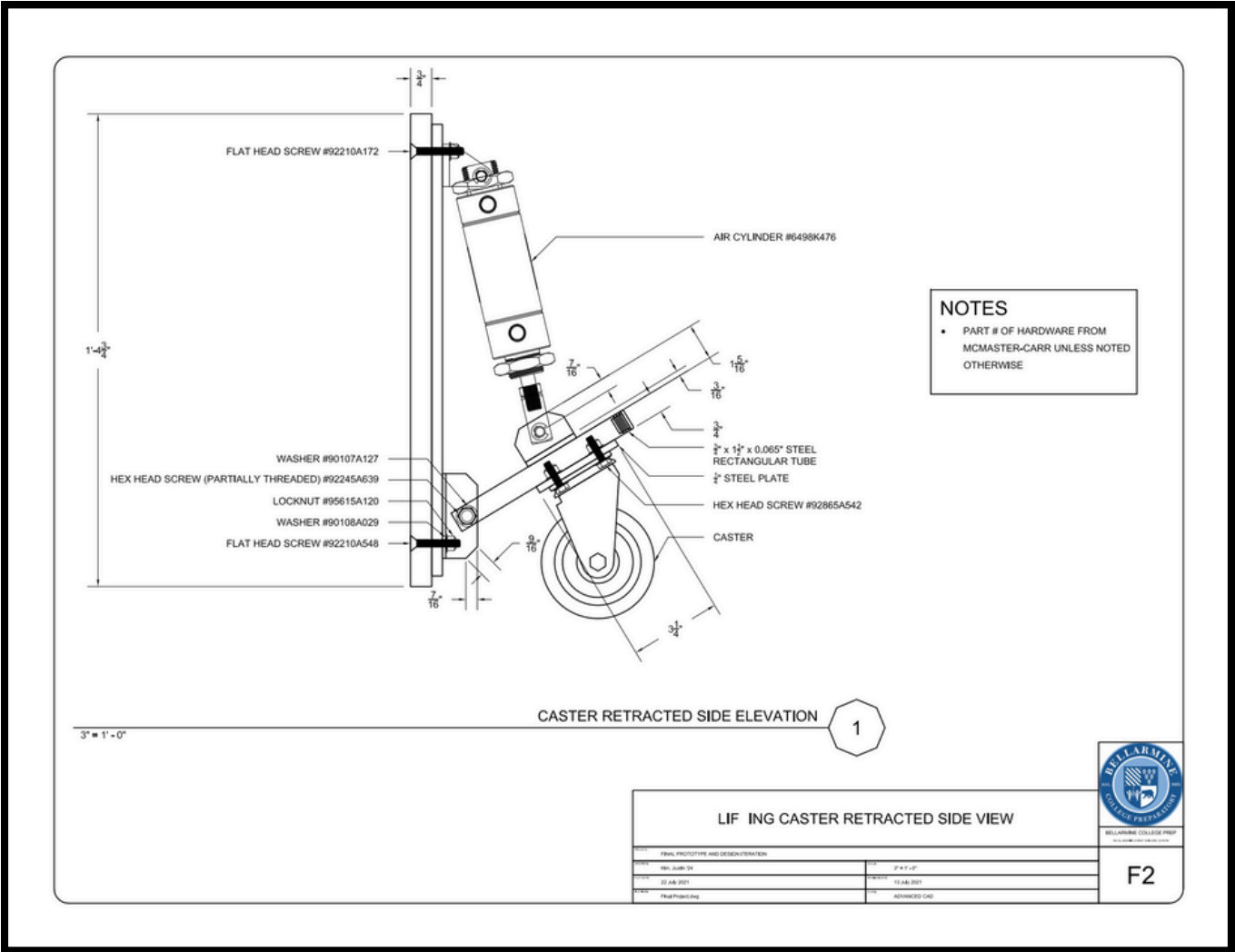
- Create a mechanism to extend and retract a caster designed to support a movable cart
- Manufacturable parts must be made out of wood
- All other parts must be commercially available

Skills and knowledge used

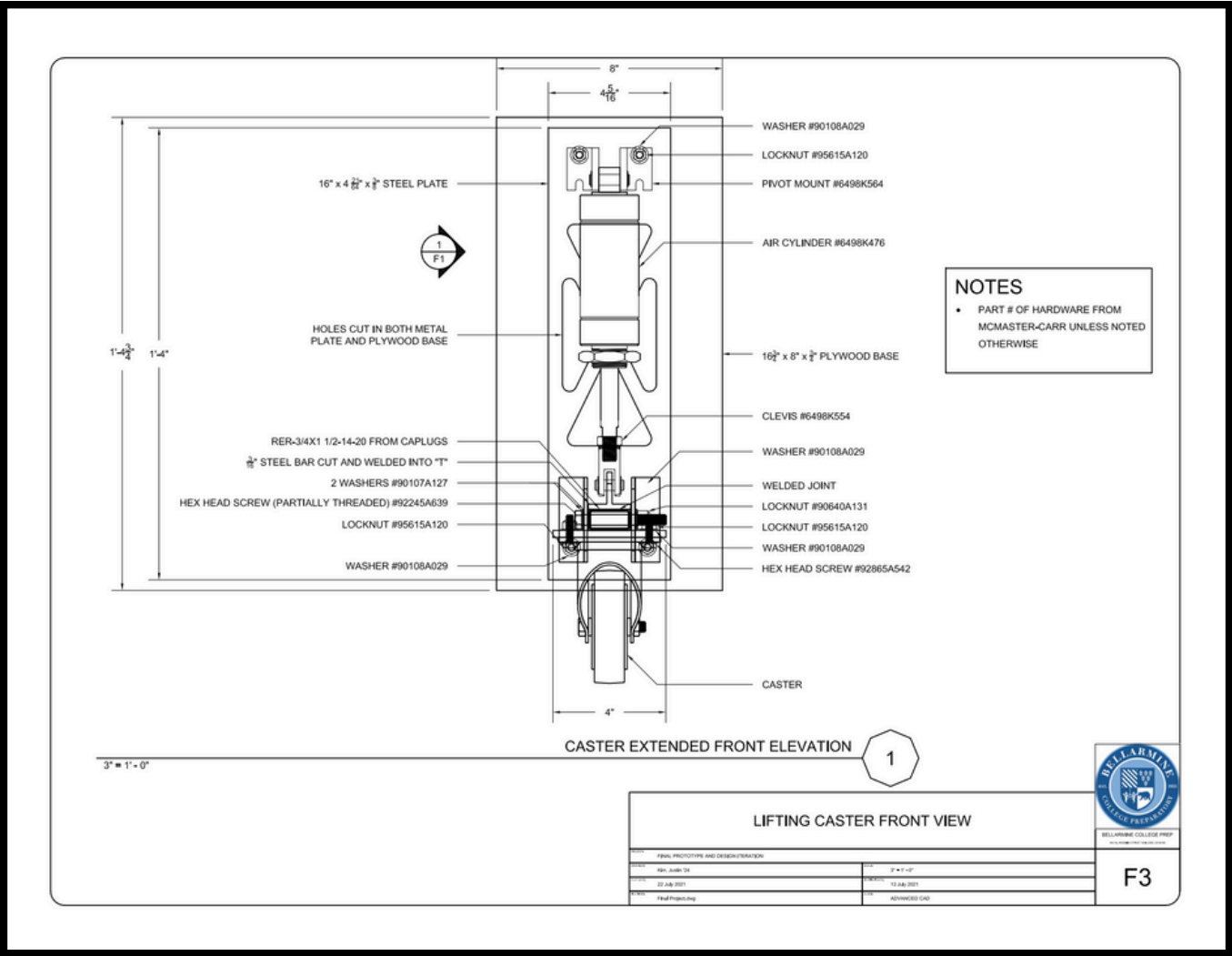
- CAD Software AutoCAD
- CAD sketches to determine pneumatic cylinder geometry
- Tolerancing
- CAD sketches to display assembly dimensions



LIFTING CASTER (AutoCAD)



LIFTING CASTER (AutoCAD)



Technical drawing of the front elevation of a metal plate. The plate is 1'-4" high and 4 5/16" wide. It features a central 'X' shape with a vertical slot through its center. The 'X' has a height of 3 1/2" and a width of 1 15/16". The vertical slot is 2 1/2" wide. There are four circular holes, one in each corner, with a diameter of 1/8". The corners are rounded with a radius of R 1/8". The bottom corners are also rounded with a radius of R 1/4". The drawing includes dimension lines and labels for all features.

METAL PLATE FRONT ELEVATION

3" = 1'-0"

1

[illegible]

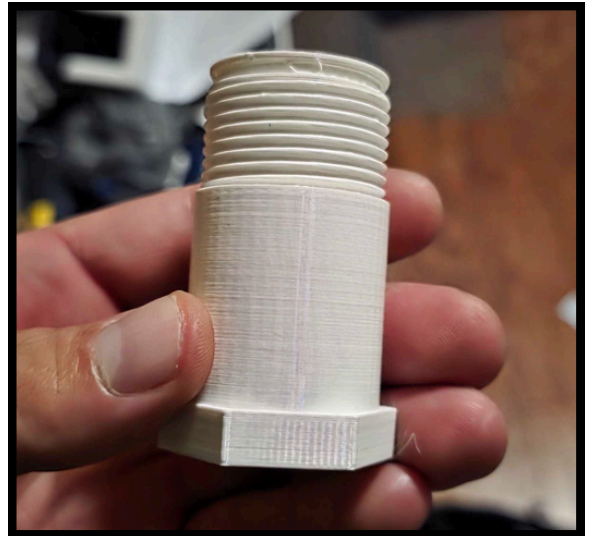
LIQUID SEALANT INJECTOR (SolidWorks)

Summary

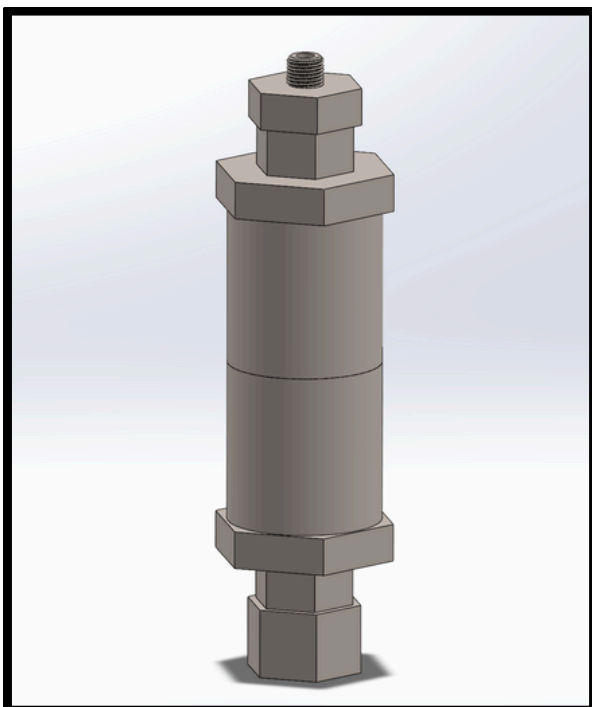
- Engineer a liquid sealant injector to pump both air and liquid sealant into bike tires for internship at Cycco
- Make hand calculations and design for pressure tank and custom ball check valves

Skills and knowledge used

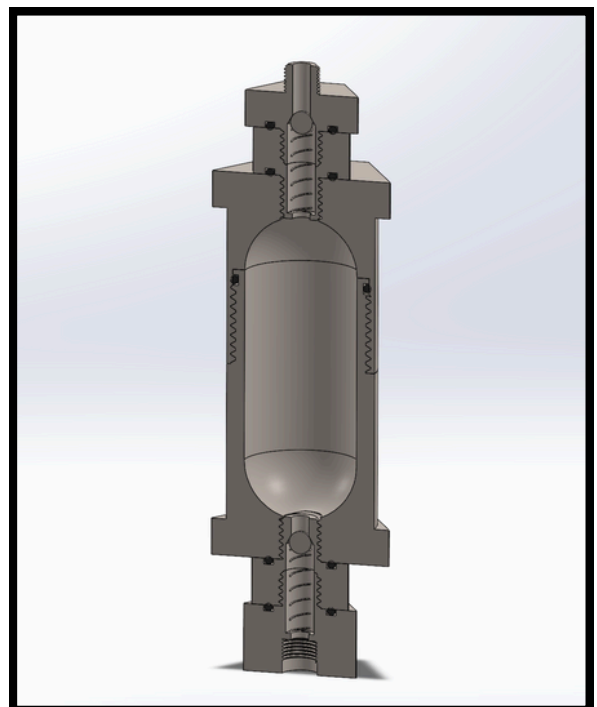
- CAD software SolidWorks
- Static and thermal FEA with SolidWorks Simulation
- Tolerancing
- O-Rings
- Newtonian Physics and Thermodynamics



3D Printed Prototype



Full CAD



Section View

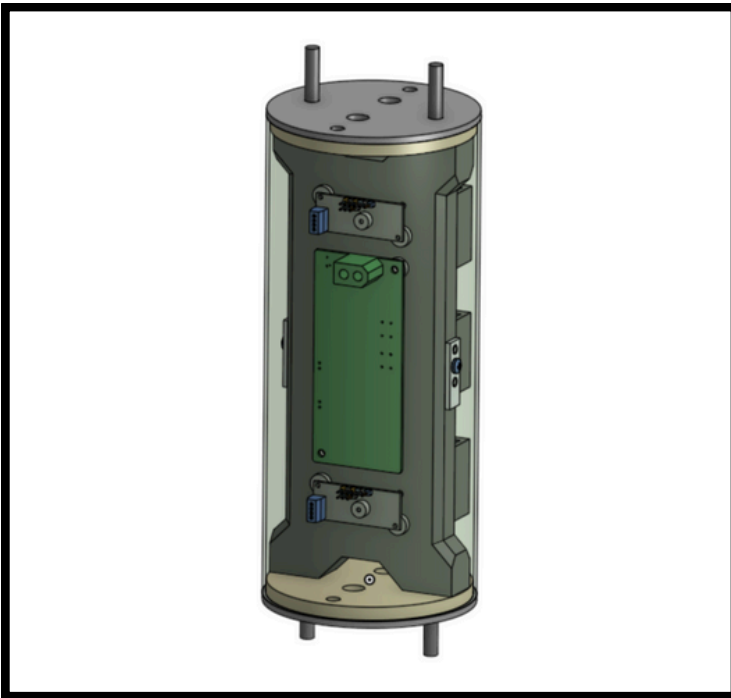
ROCKET AVIONICS BAY (Onshape)

Summary

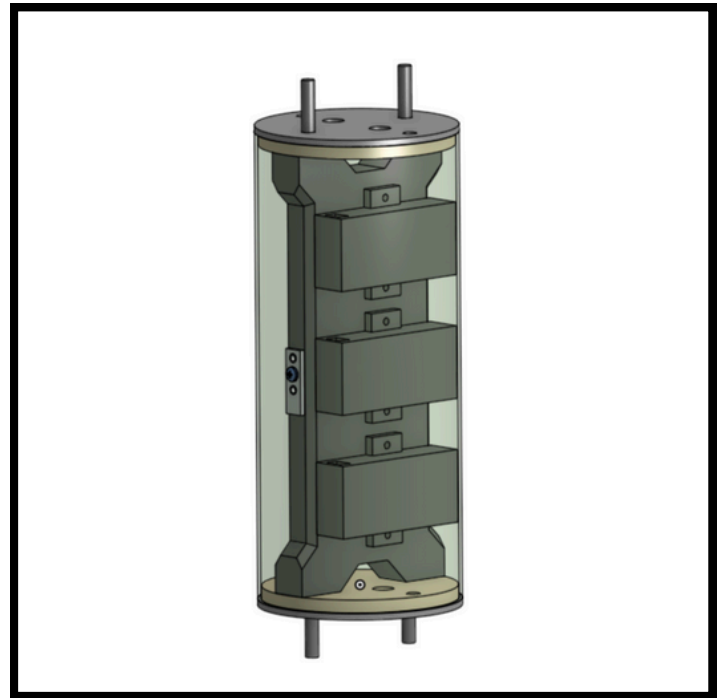
- Design a rocket's avionics bay for structural integrity, PCB integration, and battery mounts for STAR

Skills and knowledge used

- CAD software Onshape
- 3D Printing
- Tolerancing



Front View



Back View

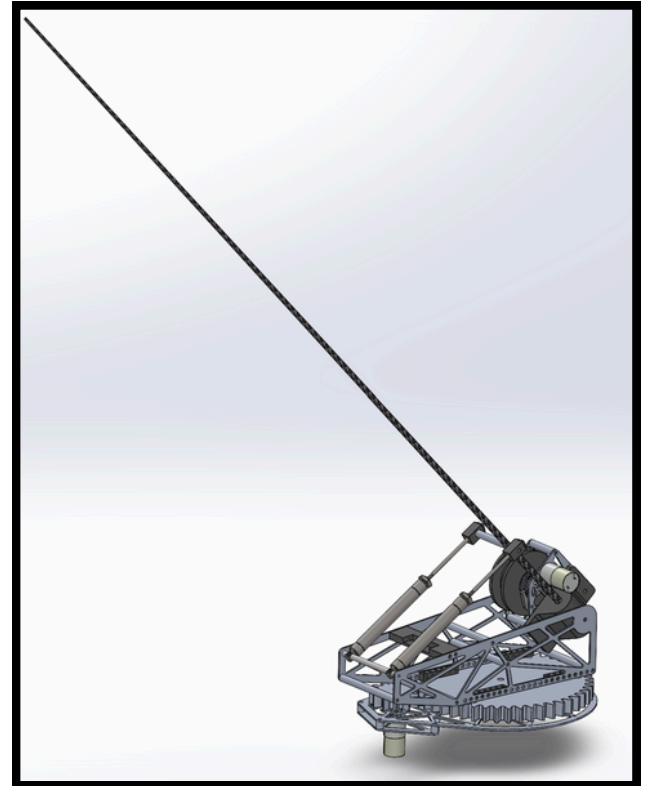
FISHING MACHINE (SolidWorks)

Summary

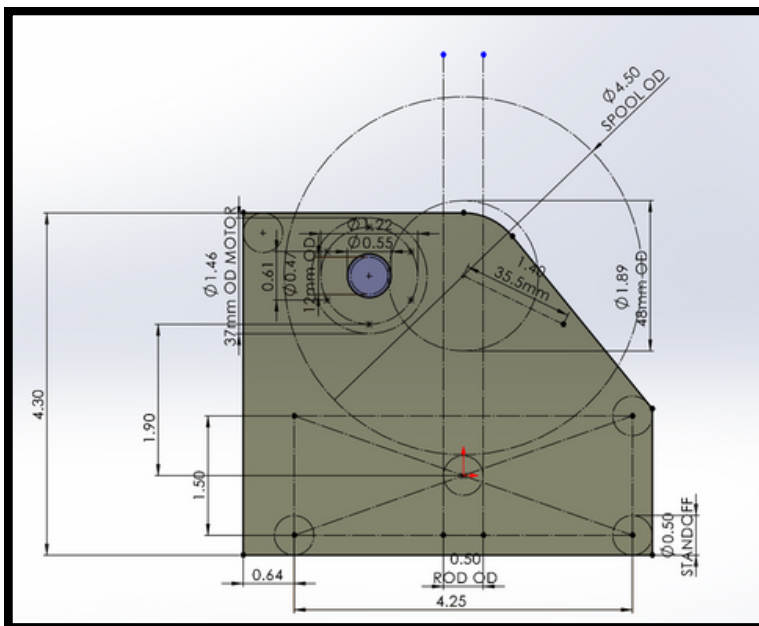
- Design an automatic fishing machine

Skills and knowledge used

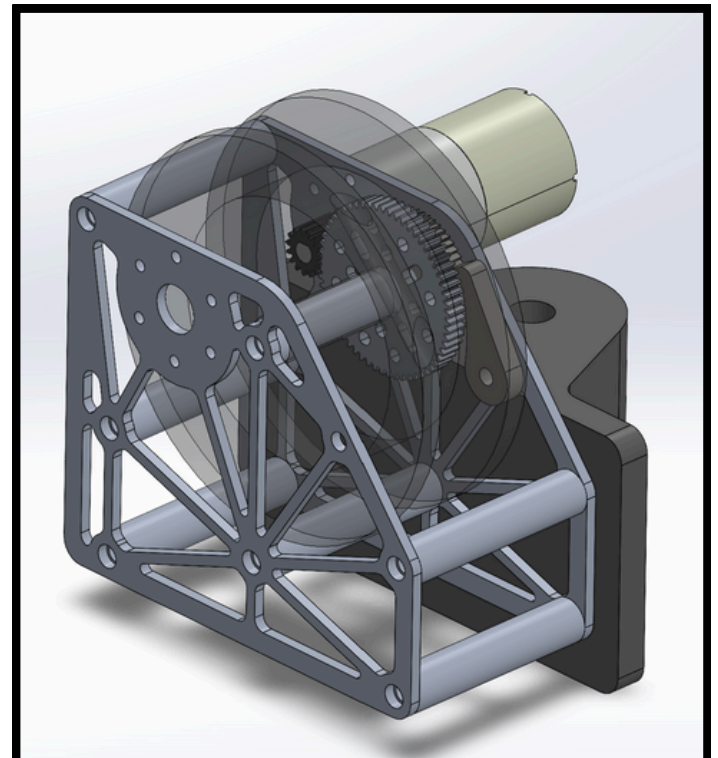
- CAD software SolidWorks
- Static FEA using SolidWorks Simulation
- CAD sketches to determine pneumatic cylinder geometry
- 3D printing
- Tolerancing
- Solid mechanics and composites
- Power transmission systems



Full Isometric View

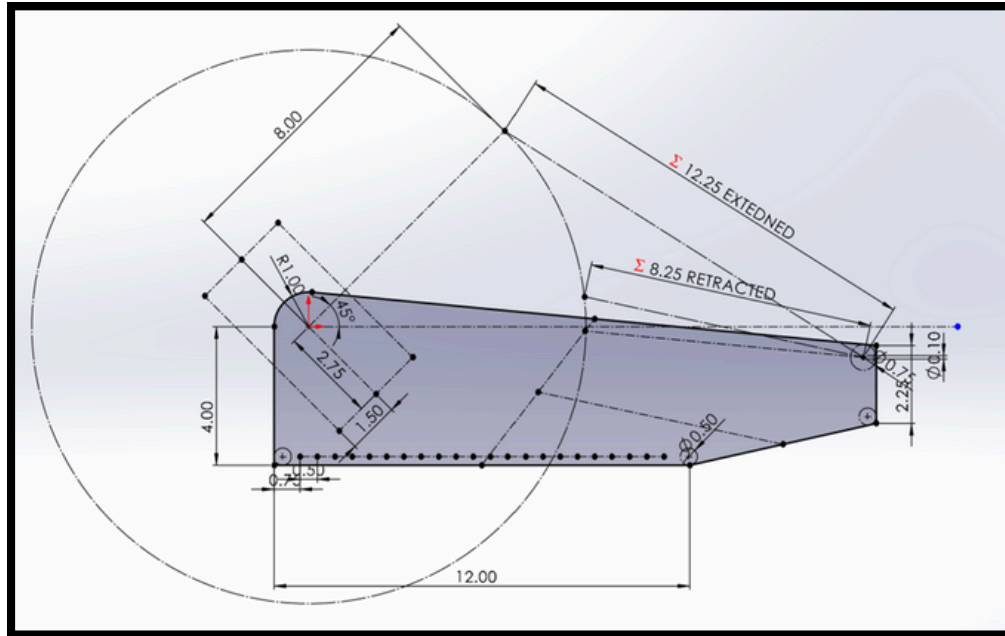


Gearbox Geometry Sketch

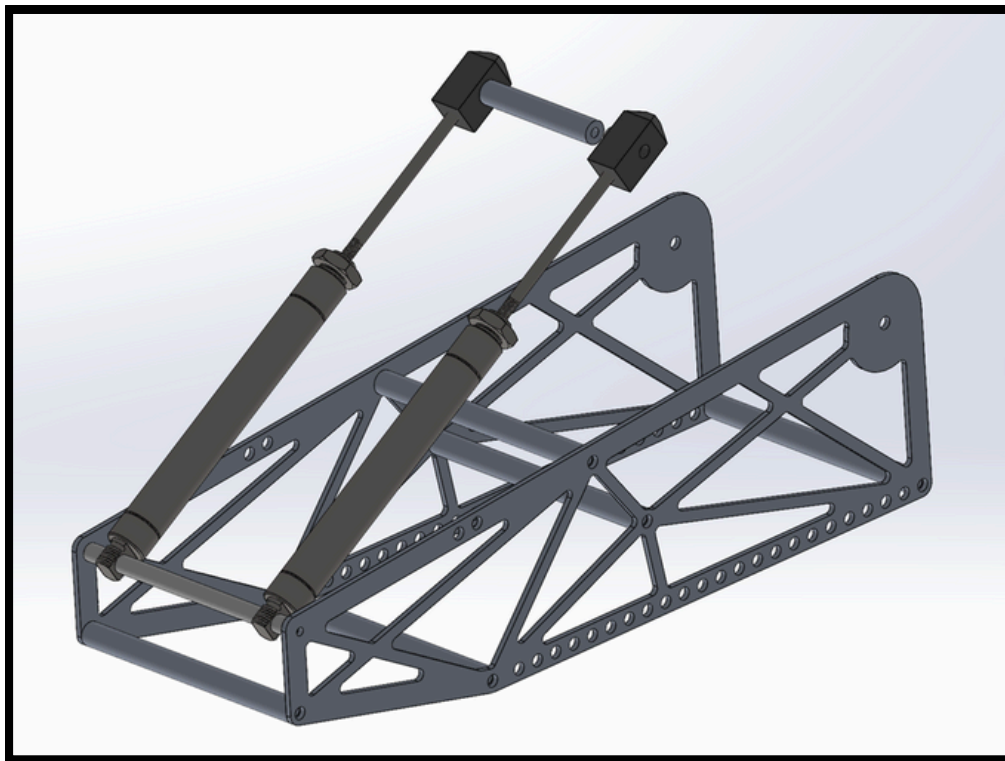


Gearbox Subassembly

FISHING MACHINE (SolidWorks)



Supporting Brace Geometry Sketch



Supporting Brace

ROBOTIC ARM (SolidWorks and Onshape)

Summary

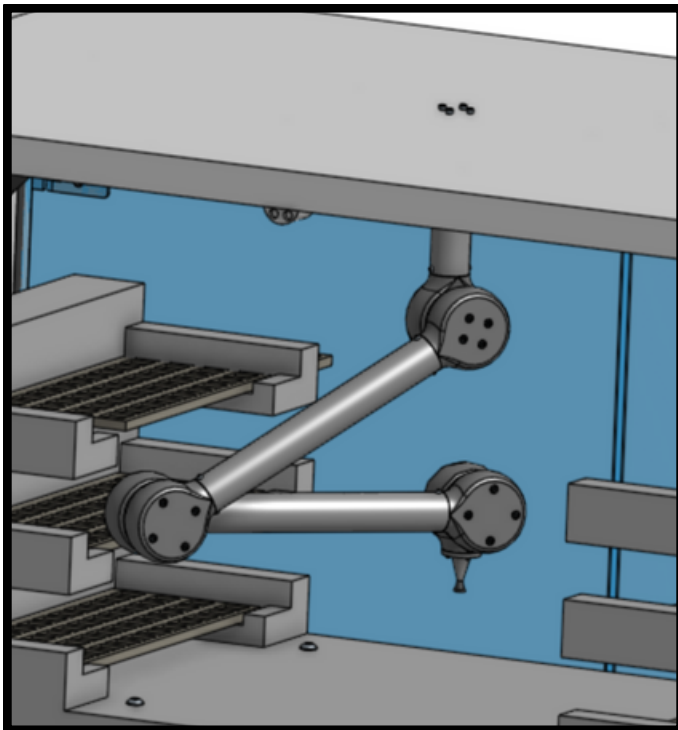
- Create a mechanism to pick up, transport, and place sensitive semiconductor materials for CADathon
- Must carry loads up to 50 lbs
- Work in a sterile vacuum environment

Skills and knowledge used

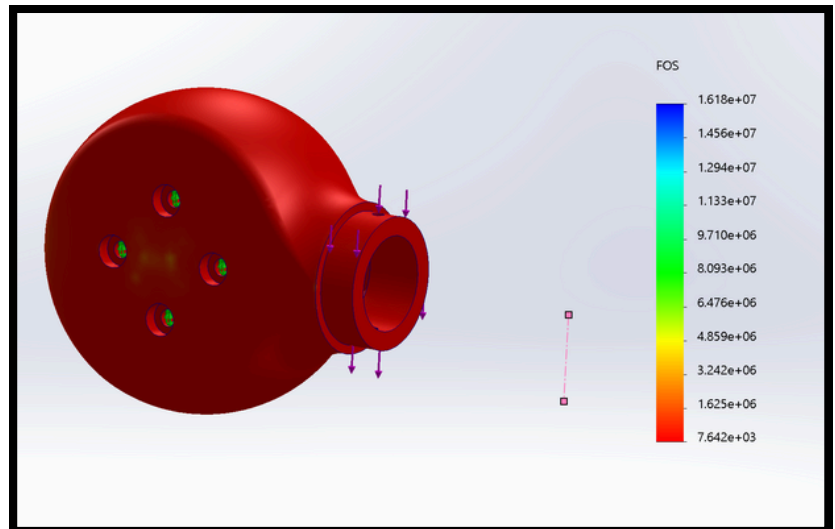
- CAD software SolidWorks and Onshape
- Static FEA using SolidWorks Simulation



Robotic Arm CAD



Robotic Arm in Sterile Environment



Robotic Arm Motor Mount FoS

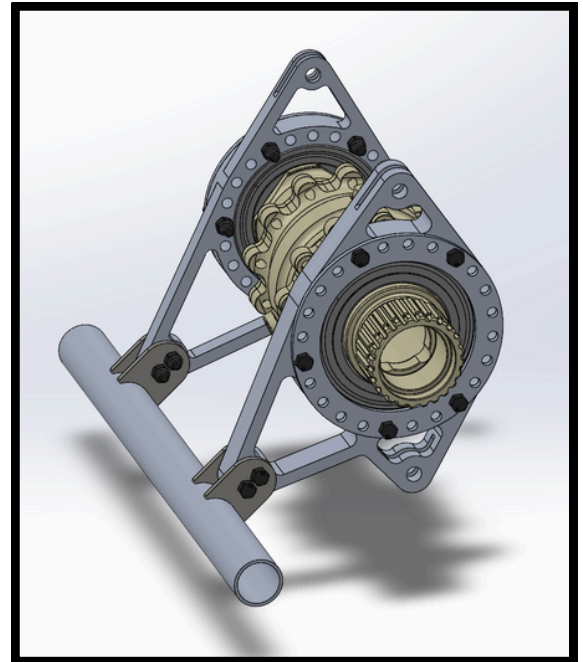
DIFFERENTIAL MOUNT (SolidWorks)

Summary

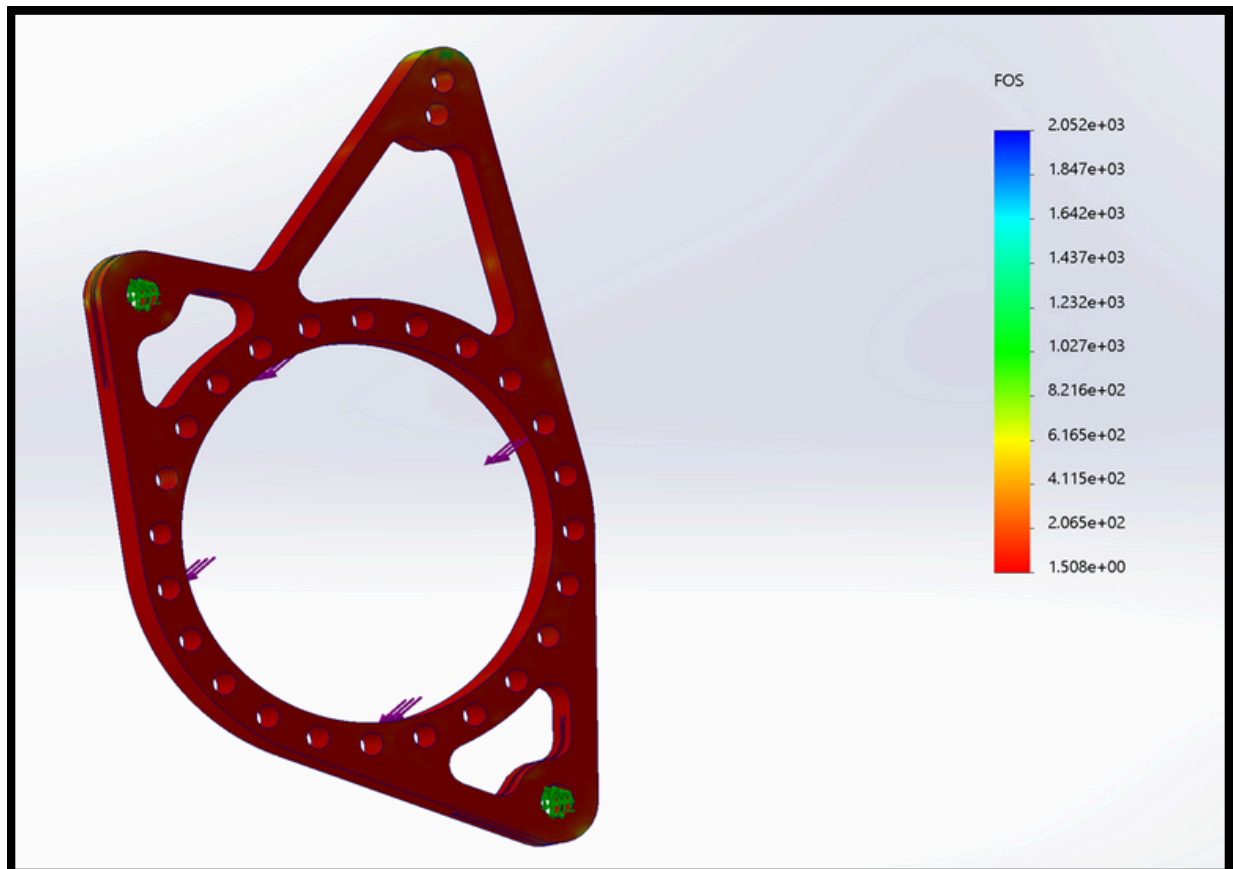
- Create a sturdy mount for an FSAE racecar differential
- Mount must secure bearings that mount on the differential
- Mount must implement a jacking bar for the car

Skills and knowledge used

- CAD software SolidWorks
- Static FEA using SolidWorks Simulation
- Tolerancing of aluminum and steel plates and its pocketing
- Knowledge of properties of different materials



Full Differential Mount CAD



Differential Mount Plate FoS

ROBOTIC SNAKE (Inventor and Onshape)

Summary

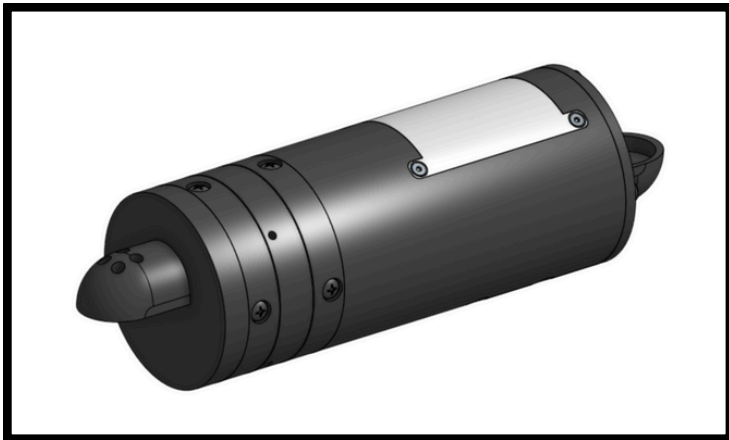
- Create a 3D printed robot that can traverse different terrain
- Inspired by NASA JPL's snake robot

Skills and knowledge used

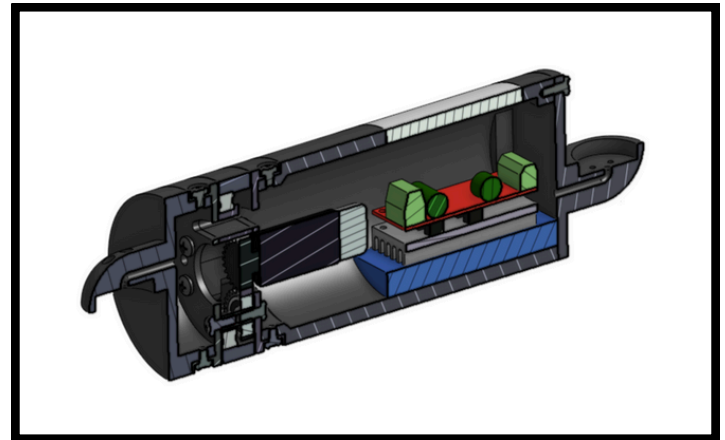
- CAD software Inventor and Onshape
- 3D printing
- Tolerancing
- Power transmission systems
- Iterative design process



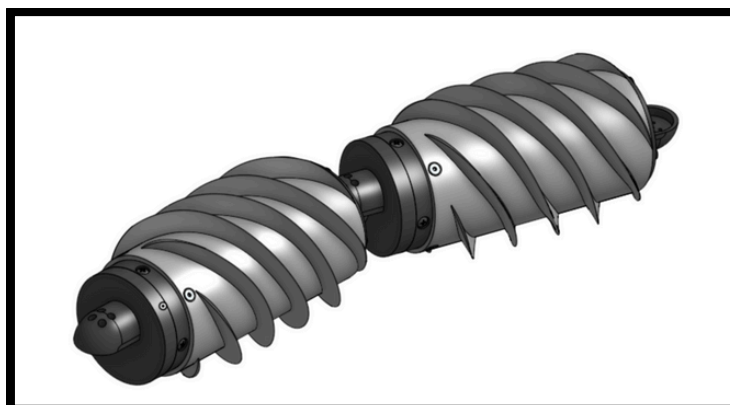
Iteration 1 3D Printed Prototype



Iteration 2 Snake Segment CAD



Iteration 2 Snake Segment Section View



Iteration 2 Full CAD

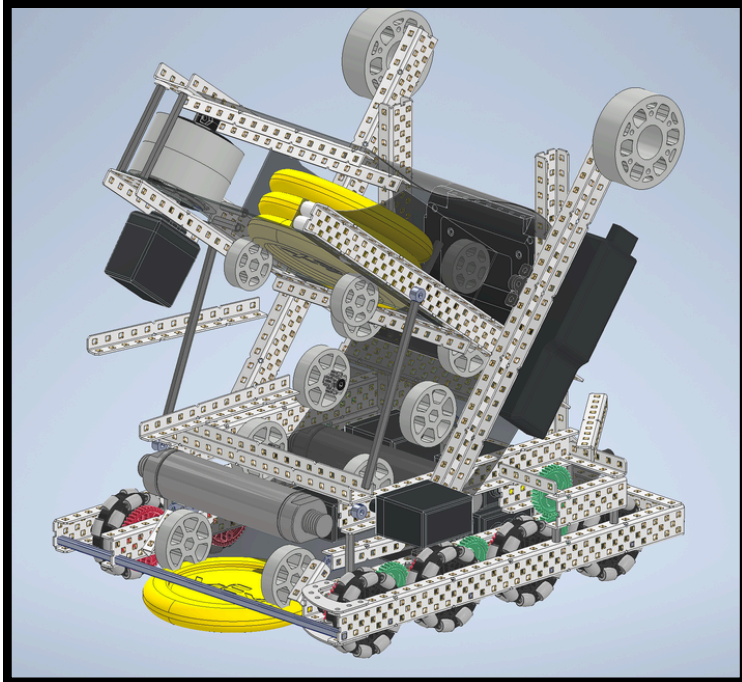
VEX COMPETITION ROBOT (Inventor)

Summary

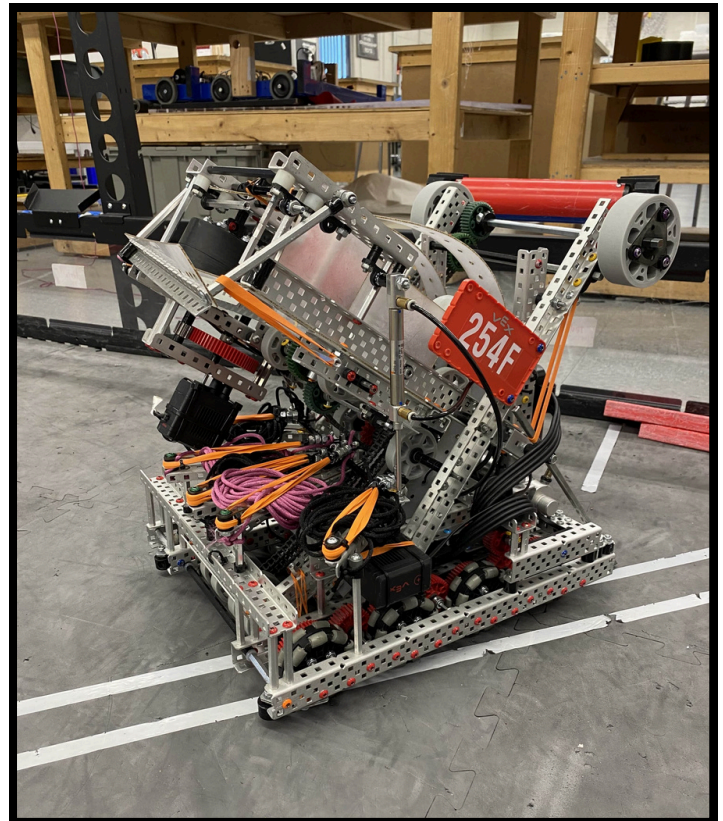
- Design and assemble a frisbee shooting robot for VEX Robotics Competition

Skills and knowledge used

- CAD software Inventor
- Mechanical assembly
- CNC Machining
- Physical testing
- Power transmission systems
- Iterative design process



Full Robot CAD



Robot Fully Assembled